

MVP (or MissVARPath): Missense Variant Pathogenicity Prediction

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1. Introduction and Problem Statement

In the era of next-generation sequencing (NGS), the primary challenge in personalized medicine has shifted from data acquisition to **variant interpretation**. Missense variants, single nucleotide substitutions that change an amino acid, are particularly difficult to classify. A vast majority are currently labeled as **Variants of Uncertain Significance (VUS)**, creating a diagnostic "bottleneck" that prevents clinicians from providing definitive treatments.

Our proposed method MVP is a machine learning framework designed to distinguish between pathogenic (disease-causing) and benign missense variants by integrating structural, evolutionary, and population-frequency data.

2. Proposed Methodology & Data Integration

The framework utilizes a multi-phased approach to ensure high-fidelity predictions while minimizing "data circularity" (the tendency of models to memorize well-known genes).

A. Data Sources

- **ClinVar:** Used for high-confidence supervised learning labels (Pathogenic vs. Benign).
- **gnomAD (v4.0):** Used to extract allele frequencies to filter out common neutral variations.
- **OpenCRAVAT:** An annotation engine used to extract hundreds of features, from different genome databases including physicochemical properties and protein structural context.

B. Feature Engineering

The model extracts orthogonal features including:

- **Conservation Scores:** phyloP, phastCons to assess evolutionary constraint.
- **Protein Dynamics:** Structural stability changes and catalytic site proximity.
- **Chemical Alterations:** Changes in molecular mass, hydrophobicity, and polarity.
- **Gene analysis:** Use of flanking strings at the point of mutation with k-mers.

3. Architecture

To achieve state-of-the-art accuracy, MVP functions as a meta-classifier that ingests scores from leading Variant Effect Predictors (VEPs):

- **AlphaMissense:** Leveraging deep learning and protein structure predictions from AlphaFold2.
- **CADD v1.7:** Utilizing protein language models and regulatory CNNs.
- **REVEL:** A forest-based ensemble optimized for rare variants.

In many cases, for VUS entries, in-silico scores are not available (alpha missense, CADD, Revel Ditto score etc.), in this case we will be investigating the sequences themselves for predicting the pathogenicity of the given missense gene.

- **k-mer:** Gene sequence will be split into 2-3 characters, and read with 1 frame shift. This will give us important information about before and after the mutated variant, allowing us to catch if the given gene became pathogenic.
- **BLAST:** Basic alignment tools are widely used in bioinformatics and can compare multiple genes or gene sites to each other to catch where the mutation happened. We will be using this tool as well to give us more insights for our prediction, merging classical bioinformatics tool and machine learning.

4. Implementation & Validation

Algorithm Selection

The core engine utilizes **XGBoost** and **LightGBM**, chosen for their ability to handle tabular biological data and capture non-linear interactions between features. To ensure the model generalizes to novel genes, we will employ **Gene-Stratified Cross-Validation**, ensuring that no variants from the same gene appear in both the training and testing sets simultaneously.

Interpretability

We will implement **SHAP (SHapley Additive exPlanations)** to provide transparent, feature-level justifications for every prediction, aligning with clinical requirements for explainable AI.

5. Expected Impact

By harmonizing disparate genomic signals into a single, high-confidence score, VariantPath aims to:

1. **Reduce VUS Rates:** Accelerate the classification of unknown variants in clinical reports.
2. **Enhance Diagnostic Yield:** Provide actionable evidence for rare disease diagnosis.
3. **Improve Clinical Precision:** Minimize false positives that lead to unnecessary medical interventions.

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